

**Alamein/Tharwa Petroleum Company  
Cairo,Egypt.**

**EAS F-3X  
AEB**

# **Conventional Frac Field Post Job Report**

**For: Alamein/Tharwa Engineers  
Date: Saturday, April 16, 2022**

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**HALLIBURTON**

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## 1.0 Well Information

### 1.1 Customer Information

|                            |                           |
|----------------------------|---------------------------|
| Customer                   | Alamein Petroleum Company |
| Well Name                  | EAS F-3X                  |
| Platform/Rig Name          | ST-12                     |
| Rig Type                   | Land - Drilling Rig       |
| Job Date                   | 16-Apr-2022               |
| Country                    | Egypt                     |
| Customer Representative    | Alamein Engineers         |
| Halliburton Representative | Mohamed Sherif            |

### 1.2 Pipe Information

| Equipment    | Top MD<br>ft | Bottom<br>MD<br>ft | Top TVD<br>ft | Bottom<br>TVD<br>ft | OD<br>in | ID<br>in | Weight<br>lb/ft |
|--------------|--------------|--------------------|---------------|---------------------|----------|----------|-----------------|
| Surface Pipe | 0.0          | 687.0              |               |                     | 4.000    | 3.000    |                 |
| Tubing       | 0.0          | 11536.7            | 0.0           | 11145.3             | 4.500    | 3.826    | 15.50           |
| Tubing       | 11536.7      | 11539.4            | 11145.3       | 11147.4             | 5.000    | 2.250    |                 |
| Tubing       | 11539.4      | 11540.8            | 11147.4       | 11148.4             | 5.000    | 2.250    |                 |
| Tubing       | 11540.8      | 11560.8            | 11148.4       | 11163.5             | 5.000    | 2.310    |                 |
| Tubing       | 11560.8      | 11562.2            | 11163.5       | 11164.6             | 5.000    | 2.250    |                 |
| Tubing       | 11562.2      | 11565.3            | 11164.6       | 11166.9             | 5.000    | 2.250    |                 |
| Tubing       | 11565.3      | 13848.7            | 11166.9       | 12888.8             | 4.500    | 3.826    | 15.50           |
| Tubing       | 13848.7      | 13851.4            | 12888.8       | 12890.8             | 5.000    | 2.250    |                 |
| Tubing       | 13851.4      | 13852.9            | 12890.8       | 12891.9             | 5.000    | 2.250    |                 |
| Tubing       | 13852.9      | 13856.9            | 12891.9       | 12895.0             | 5.000    | 2.250    |                 |
| Tubing       | 13856.9      | 13860.5            | 12895.0       | 12897.7             | 5.000    | 2.300    |                 |
| Packer       | 13860.5      | 13871.0            | 12897.7       | 12905.7             |          | 2.250    |                 |
| Tubing       | 13871.0      | 13887.9            | 12905.7       | 12918.5             | 5.000    | 2.250    |                 |
| Tubing       | 13887.9      | 13889.5            | 12918.5       | 12919.7             | 5.000    | 2.250    |                 |
| Tubing       | 13889.5      | 13950.2            | 12919.7       | 12965.5             | 3.500    | 2.992    | 9.20            |
| Tubing       | 13950.2      | 13950.8            | 12965.5       | 12965.9             | 4.500    | 2.992    |                 |
| Formation    | 14063.0      | 14085.0            | 13050.7       | 13067.4             |          |          |                 |
| Casing       | 0.0          | 14129.0            | 0.0           | 13100.6             | 7.000    | 6.184    | 29.00           |

**1.3 Perforation Intervals**

| Top MD<br>ft | Bottom<br>MD<br>ft | Top TVD<br>ft | Bottom<br>TVD<br>ft | Number<br>of<br>Shots | Perf<br>Density<br>spf | Perf<br>Phasing<br>° | Perf<br>Diameter<br>in | Perf<br>Formation |
|--------------|--------------------|---------------|---------------------|-----------------------|------------------------|----------------------|------------------------|-------------------|
| 14063.0      | 14066.0            | 13050.7       | 13053.0             | 15                    | 5                      | 60                   | 0.570                  | AEB Perfs         |
| 14070.0      | 14073.0            | 13056.0       | 13058.3             | 15                    | 5                      | 60                   | 0.570                  | AEB Perfs         |
| 14083.0      | 14085.0            | 13065.9       | 13067.4             | 10                    | 5                      | 60                   | 0.570                  | AEB Perfs         |

## 2.0 Pumping Schedule

### 2.1 Mini Frac Designed Pumping Schedule

| Stage Number | Description              | Fluid System     | Clean Volume gal | Slurry Volume gal | Rate Stage Start bpm | Rate Stage End bpm | Stage Time min |
|--------------|--------------------------|------------------|------------------|-------------------|----------------------|--------------------|----------------|
| 1            | Load Well                | HYBOR G 40#      | 1000             | 1000              | 15.0                 | 15.0               | 1.59           |
| 2            | Fluid Efficiency Test    | HYBOR G 40#      | 7970             | 7970              | 30.0                 | 30.0               | 6.33           |
| 3            | Displacement             | WATER FRAC G 40# | 9470             | 9470              | 30.0                 | 30.0               | 7.52           |
| 4            | Shut-In for FET Analysis |                  | 0                | 0                 | 0.0                  | 0.0                | 0.00           |

### 2.2 Main Frac Designed Pumping Schedule

| Stage Number | Description | Fluid System     | Clean Volume gal | Slurry Volume gal | Prop Conc Start lb/gal | Prop Conc End lb/gal | Prop Type     | Prop Mass 100*lb | Rate Stage Start bpm | Rate Stage End bpm | Stage Time min |
|--------------|-------------|------------------|------------------|-------------------|------------------------|----------------------|---------------|------------------|----------------------|--------------------|----------------|
| 1            | Pad         | HYBOR G 40#      | 19000            | 19000             |                        |                      |               |                  | 30.0                 | 30.0               | 15.08          |
| 2            | 1 ppg       | HYBOR G 40#      | 4000             | 4240              | 1.00                   | 1.00                 | 100 Mesh Sand | 40.00            | 30.0                 | 30.0               | 3.37           |
| 3            | 2 ppg       | HYBOR G 40#      | 5000             | 5600              | 2.00                   | 2.00                 | 100 Mesh Sand | 100.00           | 30.0                 | 30.0               | 4.44           |
| 4            | 2 ppg       | HYBOR G 40#      | 8500             | 9146              | 2.00                   | 2.00                 | HSP 30/50     | 170.00           | 30.0                 | 30.0               | 7.26           |
| 5            | 3 ppg       | HYBOR G 40#      | 8000             | 8888              | 3.00                   | 3.00                 | HSP 30/50     | 240.00           | 30.0                 | 30.0               | 7.05           |
| 6            | 4 ppg       | HYBOR G 40#      | 8000             | 8994              | 3.50                   | 3.50                 | HSP 30/50     | 280.00           | 30.0                 | 30.0               | 7.14           |
| 7            | 5 ppg       | HYBOR G 40#      | 7700             | 8763              | 4.00                   | 4.00                 | HSP 30/50     | 308.00           | 30.0                 | 30.0               | 6.95           |
| 8            | 6 ppg       | HYBOR G 40#      | 2500             | 2931              | 5.00                   | 5.00                 | HSP 30/50     | 125.00           | 30.0                 | 30.0               | 2.33           |
| 9            | Flush       | WATER FRAC G 40# | 8850             | 8850              | 0.00                   | 0.00                 |               |                  | 30.0                 | 30.0               | 7.02           |
| 10           | Shut-In     |                  | 0                | 0                 | 0.00                   | 0.00                 |               |                  | 0.0                  | 0.0                | 0.00           |

### 3.0 Actual Stage Summary

#### 3.1 Mini Frac Stage Summary

| Stage Number | Start Time ucts | Max Treat Pr psi | Avg Treating Pressure psi | Max Slurry Rate bpm | Avg Slurry Rate bpm | Avg Clean Rate bpm | Slurry Volume gal | Clean Volume gal | Avg Hydraulic Horsepower hp |
|--------------|-----------------|------------------|---------------------------|---------------------|---------------------|--------------------|-------------------|------------------|-----------------------------|
| 1            | 10:02:00        | 5452             | 4309                      | 14.9                | 10.1                | 10.1               | 1004              | 1004             | 1064                        |
| 2            | 10:08:30        | 8159             | 7145                      | 30.3                | 25.7                | 25.7               | 8040              | 8040             | 4494                        |
| 3            | 10:15:58        | 7381             | 6902                      | 30.4                | 28.1                | 28.1               | 9552              | 9552             | 4748                        |
| 4            | 10:24:04        | 0                | 0                         | 0.0                 | 0.0                 | 0.0                | 0                 | 0                | 0                           |

#### 3.2 Main Frac Stage Summary

| Stage Number | Start Time ucts | Max Treat Pr psi | Avg Treating Pressure psi | Max Slurry Rate bpm | Avg Slurry Rate bpm | Avg Clean Rate bpm | Slurry Volume gal | Clean Volume gal | Prop Mass 100*lb | Avg Hydraulic Horsepower hp |
|--------------|-----------------|------------------|---------------------------|---------------------|---------------------|--------------------|-------------------|------------------|------------------|-----------------------------|
| 1            | 08:55:53        | 8000             | 7132                      | 30.1                | 26.4                | 26.4               | 18797             | 18797            | 0.11             | 4615                        |
| 2            | 09:14:12        | 8216             | 8108                      | 30.0                | 29.8                | 28.4               | 4193              | 3992             | 33.63            | 5921                        |
| 3            | 09:17:33        | 8352             | 8254                      | 29.8                | 29.7                | 26.8               | 7220              | 6510             | 118.47           | 6007                        |
| 4            | 09:23:20        | 8386             | 8346                      | 29.7                | 29.7                | 27.5               | 9194              | 8516             | 175.32           | 6072                        |
| 5            | 09:30:43        | 8538             | 8426                      | 29.7                | 29.7                | 27.0               | 8702              | 7900             | 210.80           | 6135                        |
| 6            | 09:37:41        | 8825             | 8563                      | 30.3                | 29.8                | 26.3               | 11377             | 10027            | 371.67           | 6254                        |
| 7            | 09:46:46        | 8947             | 8850                      | 30.3                | 30.1                | 26.3               | 8336              | 7277             | 301.53           | 6531                        |
| 8            | 09:53:22        | 9025             | 8920                      | 30.0                | 29.3                | 25.3               | 1642              | 1418             | 65.12            | 6401                        |
| 9            | 09:54:42        | 10821            | 9882                      | 30.2                | 30.0                | 30.0               | 8860              | 8860             | 0.00             | 7258                        |
| 10           | 10:01:47        | 0                | 0                         | 0.0                 | 0.0                 | 0.0                | 0                 | 0                | 0.00             | 0                           |

## 4.0 Performance Highlights

### 4.1 Mini Frac Job Summary

|                              |                    |      |
|------------------------------|--------------------|------|
| <b>Start Time</b>            | 16-Apr-22 08:58:07 | ucts |
| <b>End Time</b>              | 16-Apr-22 17:11:04 | ucts |
| <b>Pump Time</b>             | 17.90              | min  |
| <b>Max Treating Pressure</b> | 8159               | psi  |
| <b>Avg Treating Pressure</b> | 6660               | psi  |
| <b>Max Slurry Rate</b>       | 30.4               | bpm  |
| <b>Avg Slurry Rate</b>       | 24.7               | bpm  |
| <b>Slurry Volume</b>         | 18596              | gal  |

### 4.2 Main Frac Job Summary

|                              |                    |        |
|------------------------------|--------------------|--------|
| <b>Start Time</b>            | 18-Apr-22 06:54:55 | ucts   |
| <b>End Time</b>              | 18-Apr-22 10:20:00 | ucts   |
| <b>Pump Time</b>             | 64.32              | min    |
| <b>Max Treating Pressure</b> | 10821              | psi    |
| <b>Avg Treating Pressure</b> | 8276               | psi    |
| <b>Max Slurry Rate</b>       | 30.3               | bpm    |
| <b>Avg Slurry Rate</b>       | 28.9               | bpm    |
| <b>Slurry Volume</b>         | 78322              | gal    |
| <b>Max Prop Conc</b>         | 5.30               | lb/gal |
| <b>Prop Mass</b>             | 1254.00            | 100*lb |

### 4.3 Volumes Summary

|                                           |         |        |
|-------------------------------------------|---------|--------|
| <b>Mini Frac (Hybor G 40#)</b>            | 9044    | gal    |
| <b>Mini Frac (Linear Gel 40#)</b>         | 9552    | gal    |
| <b>Main Frac (Hybor G 40#)</b>            | 64473   | gal    |
| <b>Main Frac (Linear Gel 40#)</b>         | 8860    | gal    |
| <b>Linear Gel 35# (Left for Cleanout)</b> | 46944   | gal    |
| <b>100 mesh sand</b>                      | 132.00  | 100*lb |
| <b>HSP 30/50</b>                          | 1122.00 | 100*lb |

Disclaimer: The average and maximum values (except volumes and bottom hole values) are based on the start and end averaging times.



## 5.0 Quality Assurance

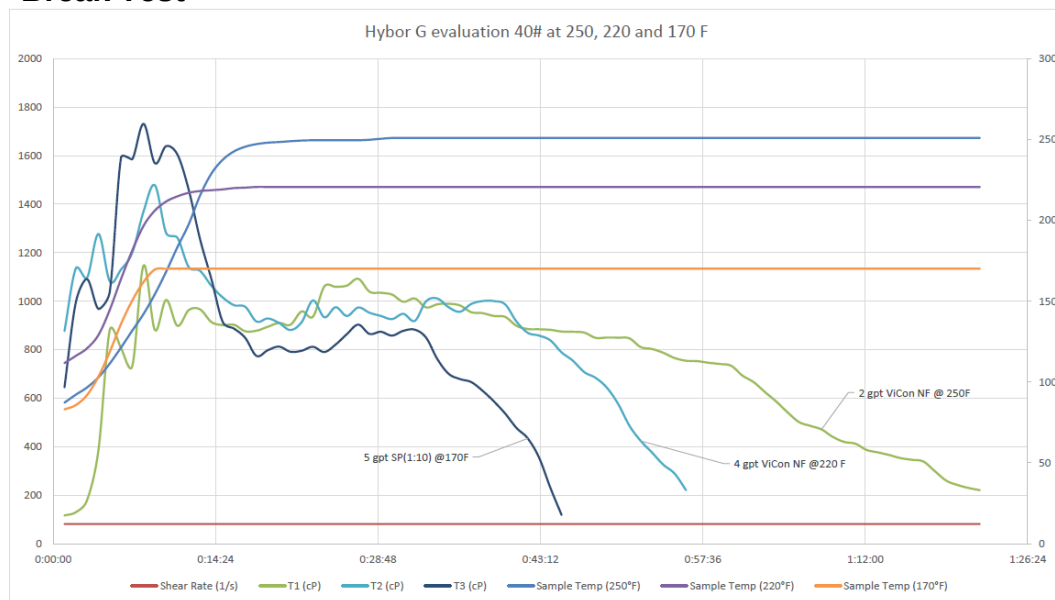
### 5.1 Fluid Analysis Tests

| Tank | Fluid Name   | Fluid Density lb/gal | Test Temp °F | pH   | Iron Conc mg/L | Phosphates Conc mg/L | Sulfate Conc mg/L |
|------|--------------|----------------------|--------------|------|----------------|----------------------|-------------------|
| 1    | 4% KCL Water | 8.54                 | 65.0         | 7.35 | 1.30           | 1.30                 | 190.00            |
| 2    | 4% KCL Water | 8.54                 | 65.0         | 7.45 | 1.10           | 1.00                 | 85.00             |
| 3    | 4% KCL Water | 8.54                 | 65.0         | 7.25 | 3.00           | 2.00                 | 50.00             |
| 4    | 4% KCL Water | 8.54                 | 65.0         | 7.65 | 1.00           | 3.00                 | 65.00             |
| 5    | 4% KCL Water | 8.54                 | 64.0         | 7.58 | 1.30           | 2.00                 | 45.00             |

### 5.2 Crosslink Test

| Fluid Name  | X-Linker | XLinker Conc gal/Mgal | Accelerator | Accelerator Conc gal/Mgal | Fluid Temp °F | Xlinker pH | Base Fluid Visc cp | Vortex min | Lip min | Clean min |
|-------------|----------|-----------------------|-------------|---------------------------|---------------|------------|--------------------|------------|---------|-----------|
| HYBOR G 40# | CL-28M   | 1.80                  | CL-31       | 0.40                      | 75.0          | 11.00      | 43.00              | 1.00       | 2.30    | 2.40      |

### 5.3 Break Test



## 5.4 Sand Sieve Tests

### Sieve Test

|                        |           |        |
|------------------------|-----------|--------|
| <b>Name</b>            | HSP 30/50 |        |
| <b>Description</b>     | HSP 30/50 |        |
| <b>Absolute Volume</b> | 0.03800   | gal/lb |
| <b>Sieve Size</b>      | 30/50     |        |
| <b>Sieve Type</b>      | Standard  |        |
| <b>Sample Mass</b>     | 100.000   | g      |
| <b>Passed Test</b>     | Yes       |        |

| Sieve Size | Amount Retained<br>g | Percent<br>% |
|------------|----------------------|--------------|
| #20        | 0.000                | 0.0          |
| #30        | 0.280                | 0.3          |
| #35        | 89.100               | 89.1         |
| #40        | 10.300               | 10.3         |
| #45        | 0.520                | 0.5          |
| #50        | 0.040                | 0.0          |
| Pan        | 0.000                | 0.0          |
| Total      | 100.240              | 100.2        |

## 5.5 Real-Time Job QA/QC

| Time<br>(Hr:Min) | Stage Description | Fluid Type  | Crosslink Time<br>(Mins:Sec) |
|------------------|-------------------|-------------|------------------------------|
| 9:00             | PAD               | HYBOR G #40 | 2:30                         |
| 9:10             | PAD               | HYBOR G #40 | 2:35                         |
| 9:15             | 1 ppg             | HYBOR G #40 | 2:35                         |
| 9:20             | 2 ppg             | HYBOR G #40 | 2:35                         |
| 9:26             | 2 ppg             | HYBOR G #40 | 2:35                         |
| 9:35             | 3 ppg             | HYBOR G #40 | 2:32                         |
| 9:46             | 3.5 ppg           | HYBOR G #40 | 2:30                         |
| 9:50             | 4 ppg             | HYBOR G #40 | 2:32                         |
| 9:55             | 5 ppg             | HYBOR G #40 | 2:35                         |

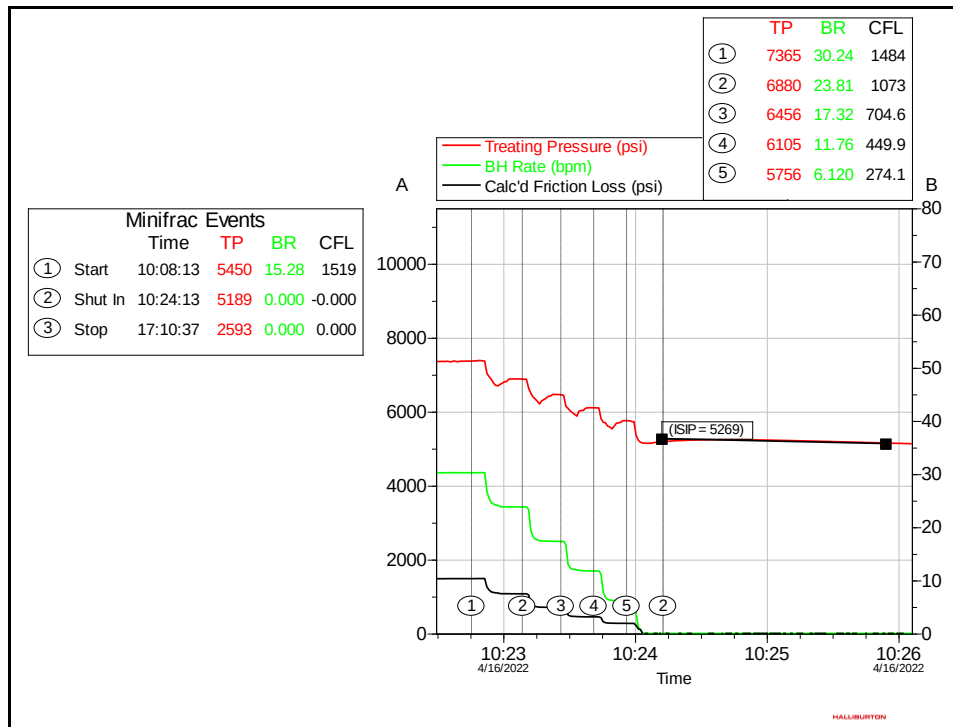
## 6.0 Attachments

### 6.1 Pumping Diagnostic Analysis Toolkit (PDAT)

#### ➤ General Information

Pressure Used: Surface  
Treating Pressure Source: Logical - Treating Pressure (psi)  
Treating Rate Source: Logical - BH Rate (bpm)  
Average Pump Rate: 30.28 bpm  
Pump Time: 14.1016 min

#### ➤ Job Data



## ➤ Analysis

|                    |            |
|--------------------|------------|
| KANE-Pad:          | 2.04 %     |
| N-Pad:             | 7.04 %     |
| S-Pad:             | 7.70 %     |
| Bottomhole ISIP:   | 11222psi   |
| Fracture Gradient: | 0.86psi/ft |

## Graph Table

| Bottom Hole<br>ISIP<br>(psi) |
|------------------------------|
| 11222                        |

**Note: Due to very slow leakoff, closure pressure was not observed on surface and will be validated once bottomhole gauges data is retrieved.**

## ➤ Step Down Analysis Input

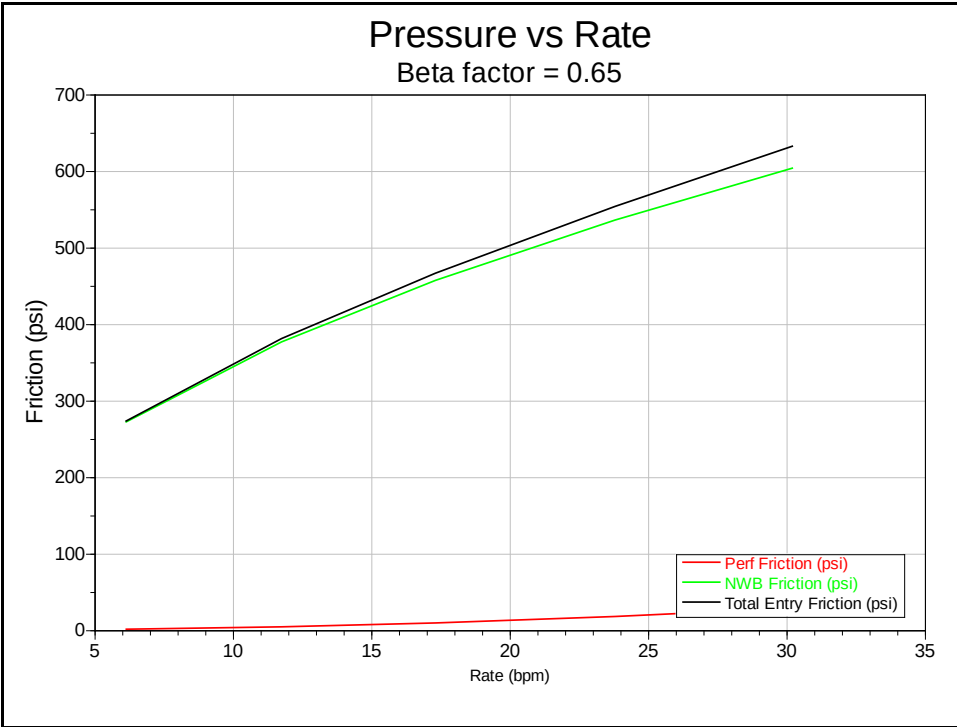
ISIP: 5269 psi  
Fluid Density: 8.54 lb/gal  
Entry Hole Size: 0.570 in  
Number of Perforations: 40  
Fluid type: Linear Gel

## ➤ Step Down Analysis Results

Beta Factor: 0.65  
Pipe Friction: 1484 psi  
Entry Friction: 633 psi  
 $K_{\text{Entry}}$ : 70.54  
Perf Friction: 29 psi  
 $K_{\text{Perf}}$ : 0.03  
NWB Friction: 604 psi  
 $K_{\text{NWB}}$ : 109.82

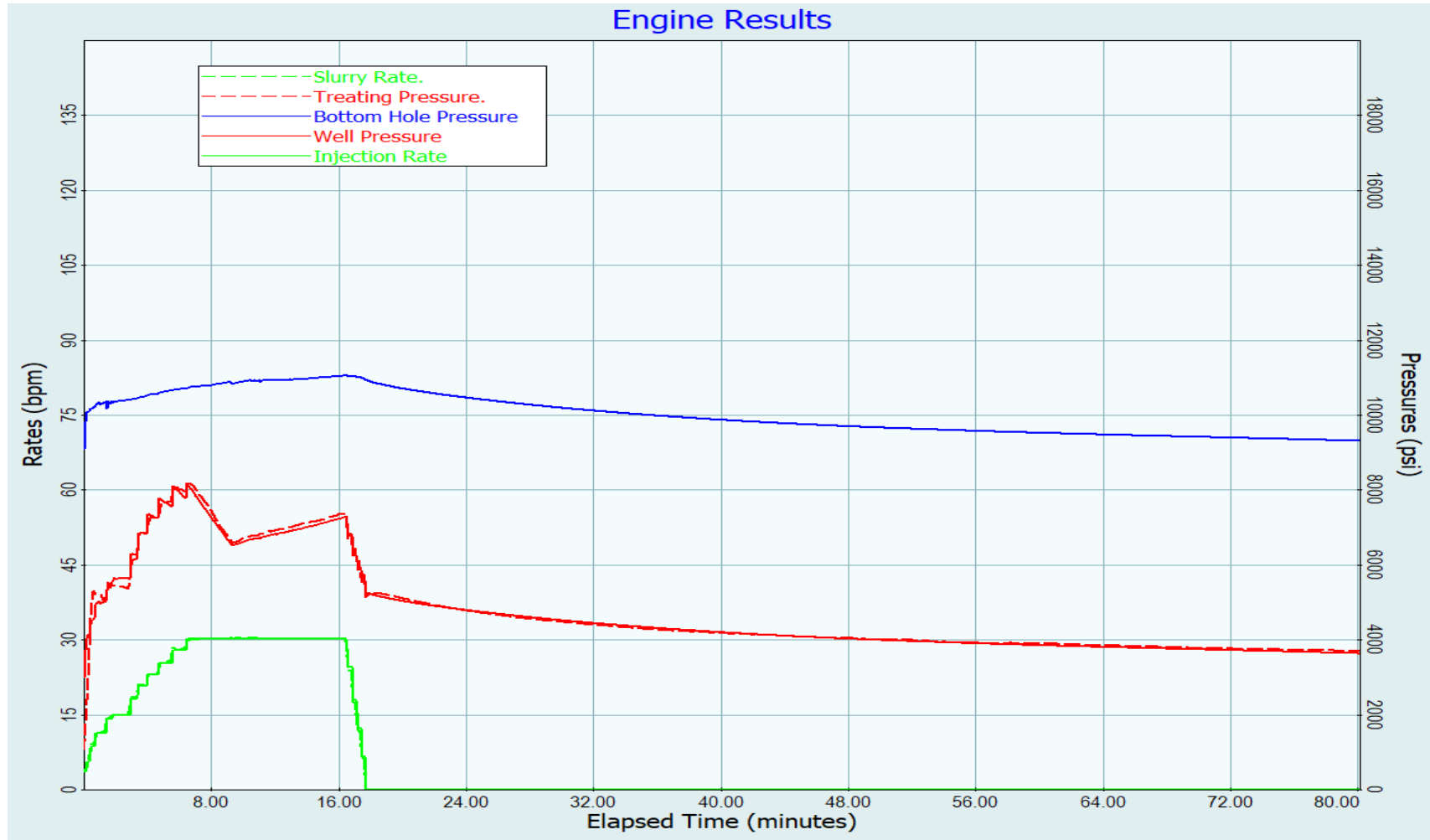
## ➤ Step Down Analysis Data

| Step Time            | Step Pressure<br>psi | Step Rate<br>bpm | Pipe Friction<br>psi | Entry Friction<br>psi | Perf Friction<br>psi | NWB Friction<br>psi |
|----------------------|----------------------|------------------|----------------------|-----------------------|----------------------|---------------------|
| 16-Apr-2022 10:22:45 | 7365                 | 30.24            | 1484                 | 633                   | 29                   | 604                 |
| 10:23:08             | 6880                 | 23.81            | 1073                 | 554                   | 18                   | 536                 |
| 10:23:26             | 6456                 | 17.32            | 705                  | 466                   | 9                    | 457                 |
| 10:23:41             | 6105                 | 11.76            | 450                  | 381                   | 4                    | 377                 |
| 10:23:56             | 5756                 | 6.12             | 274                  | 273                   | 1                    | 272                 |

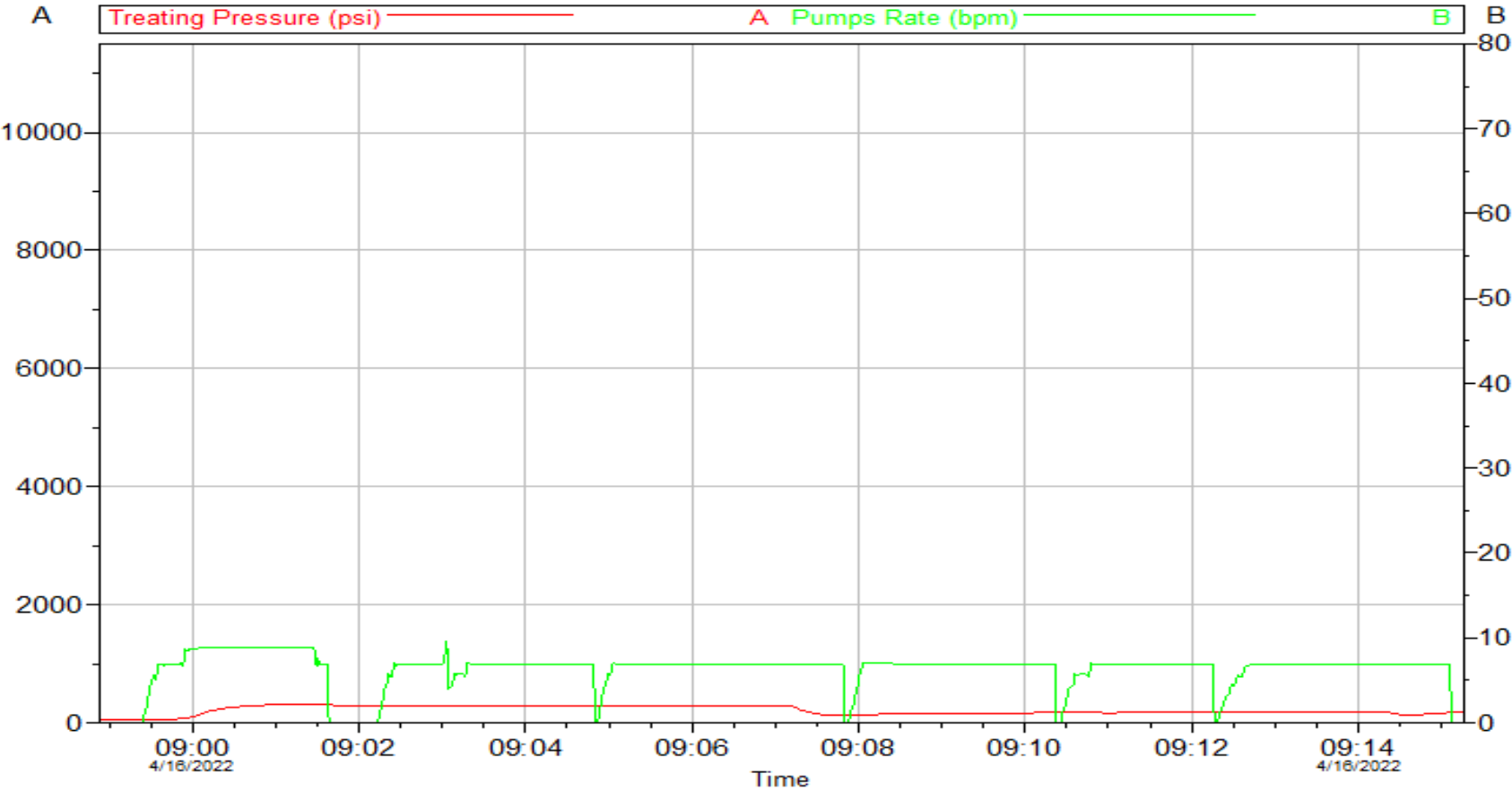


## 6.2 GOHFER Model Match

### ➤ Mini Frac Match

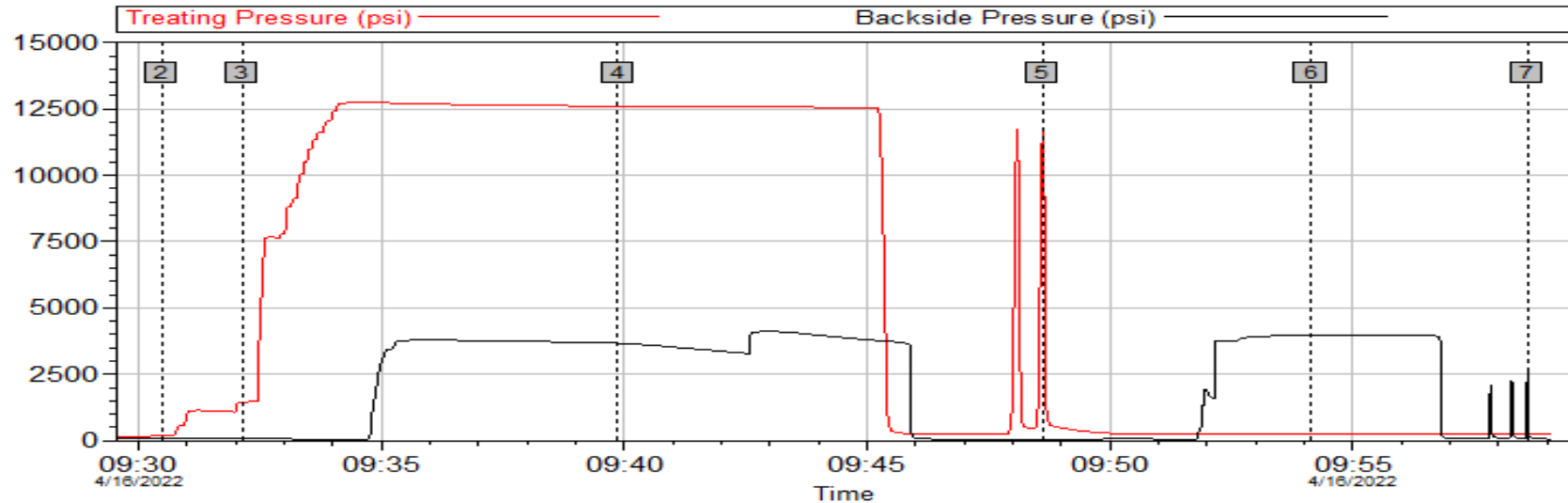


6.3 Mini Frac High Pressure Pumps Priming





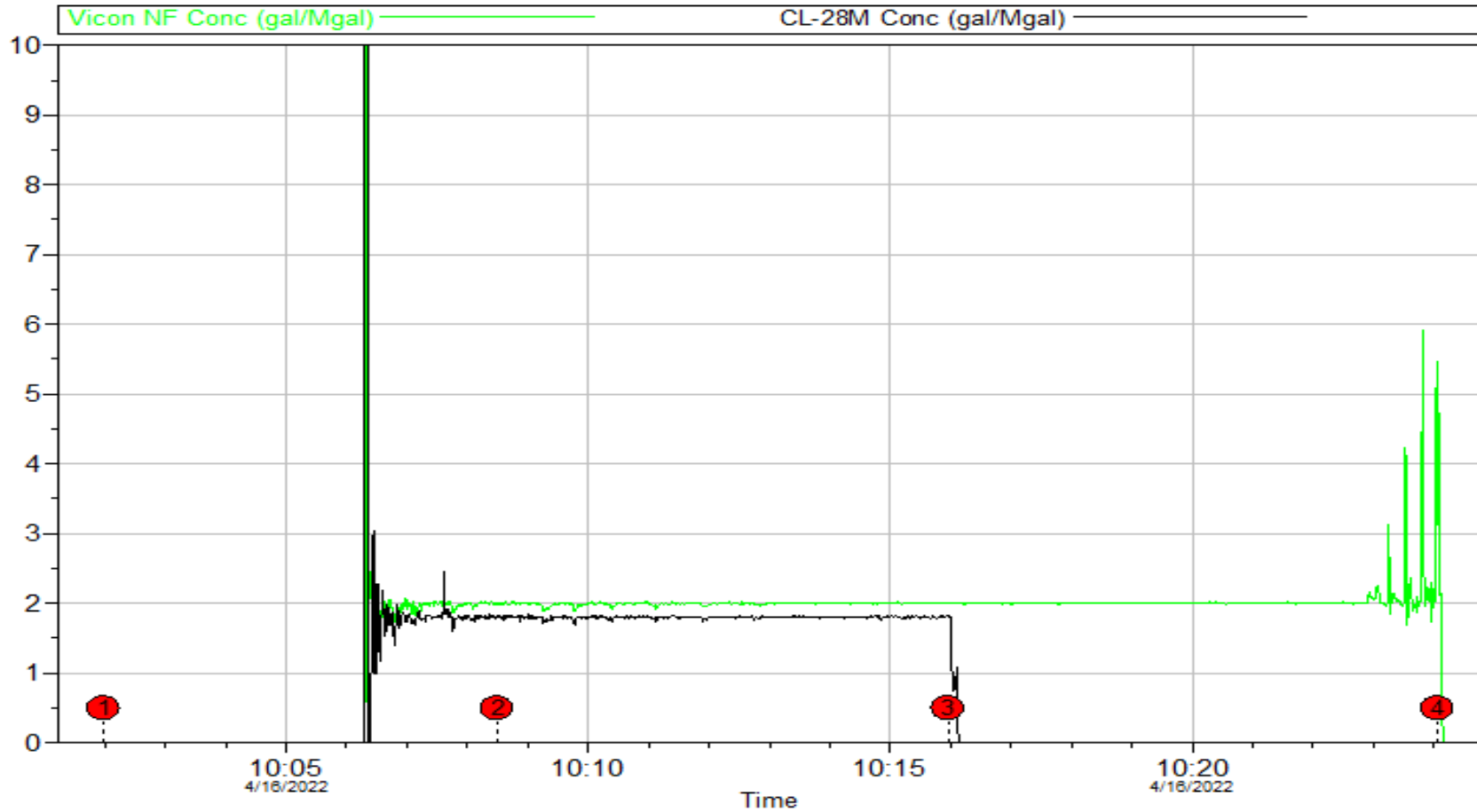
## 6.4 Mini Frac Pressure Test Summary



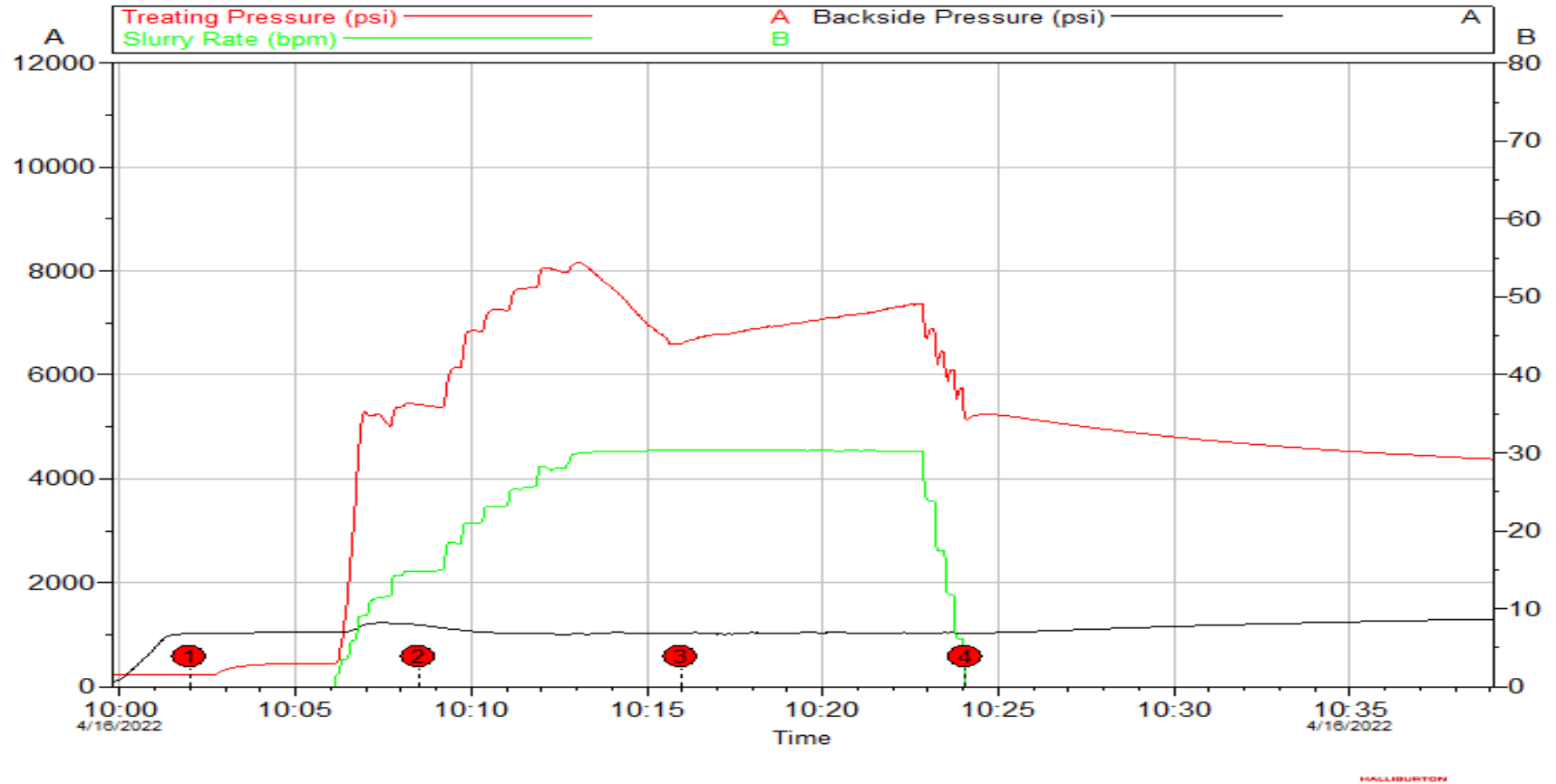
| Global Event Log                      |             |             |
|---------------------------------------|-------------|-------------|
| Intersection                          | Treating Pr | Backside Pr |
| 2 Global Kickout System Pressure Test | 180.2       | 102.1       |
| 3 Local Kickout System Pressure Test  | 1444        | 79.30       |
| 4 Treating Lines Pressure Test        | 12598       | 3672        |
| 5 Treating Line PRV                   | 11575       | 15.97       |
| 6 Backside Line Pressure Test         | 230.0       | 3949        |
| 7 Backside Line PRV                   | 229.7       | 2619        |

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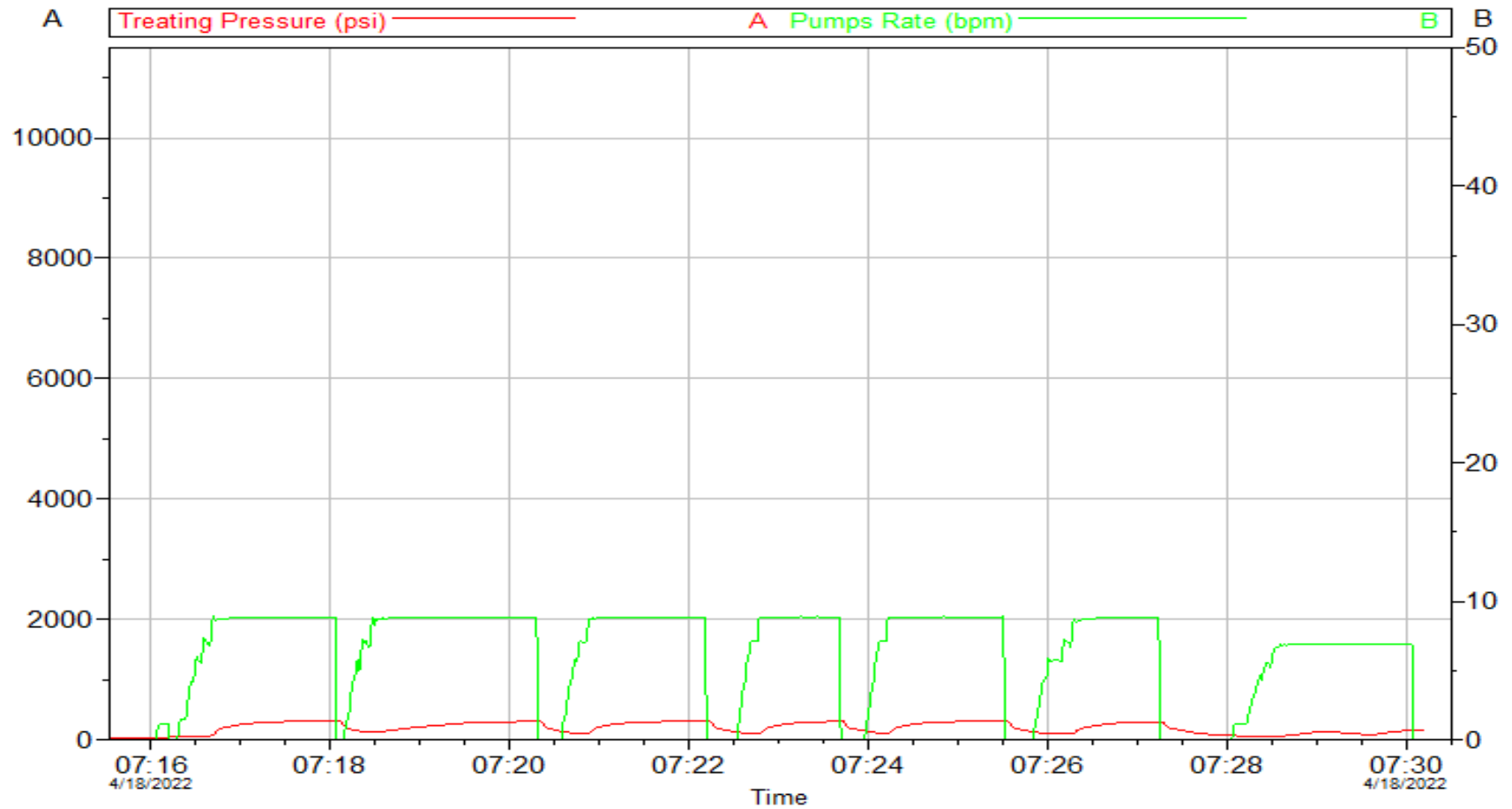
## 6.5 Mini Frac Chemical Additives Job Summary



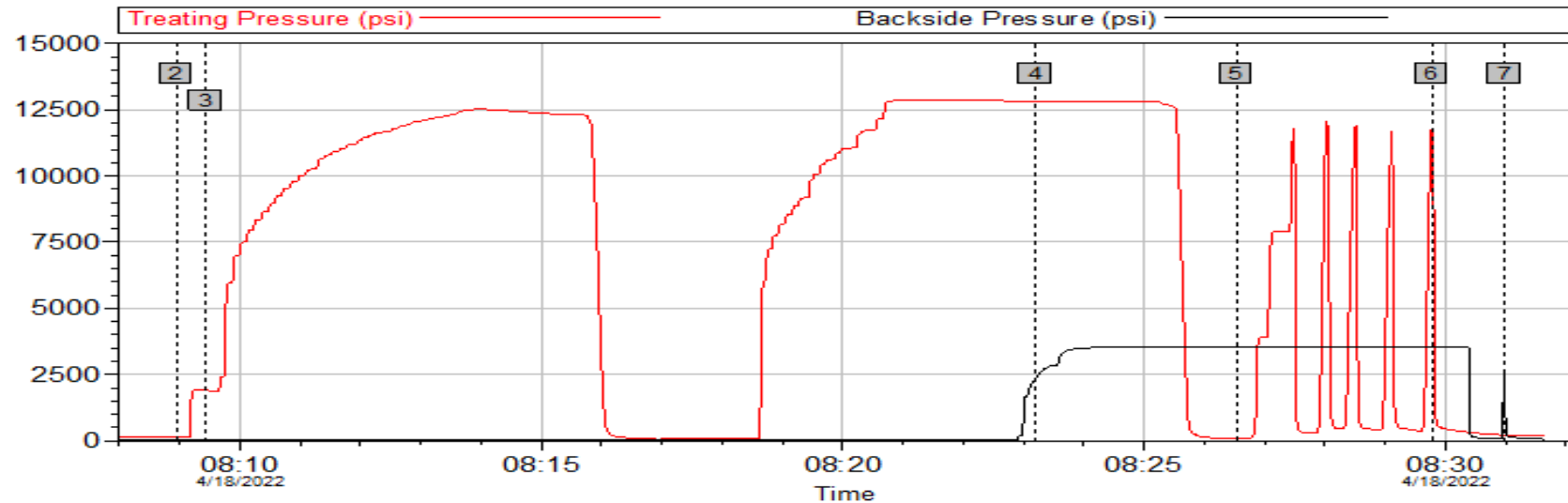
## 6.6 Mini Frac Job Summary



## 6.7 Main Frac High Pressure Pumps Priming



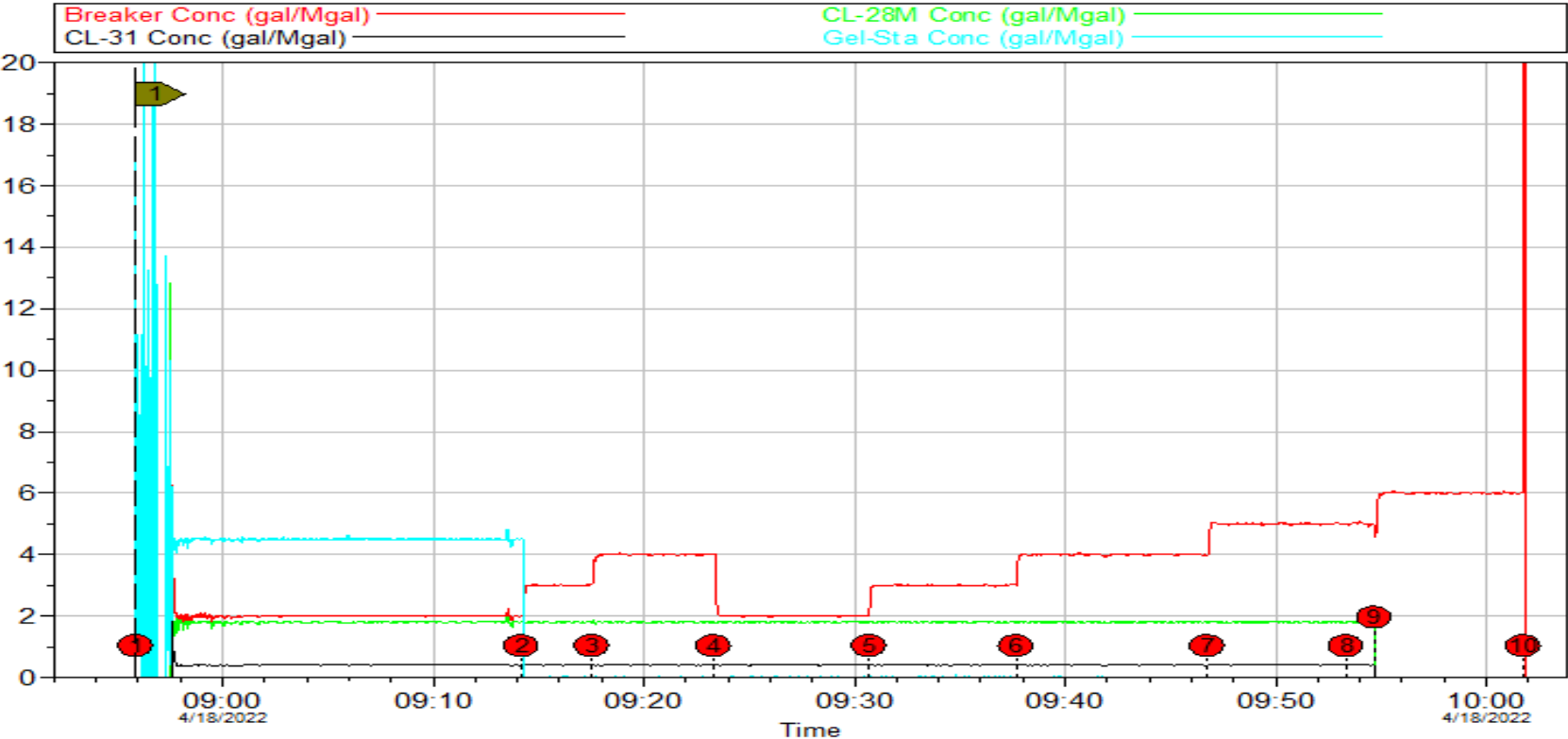
## 6.8 Main Frac Pressure Test Summary



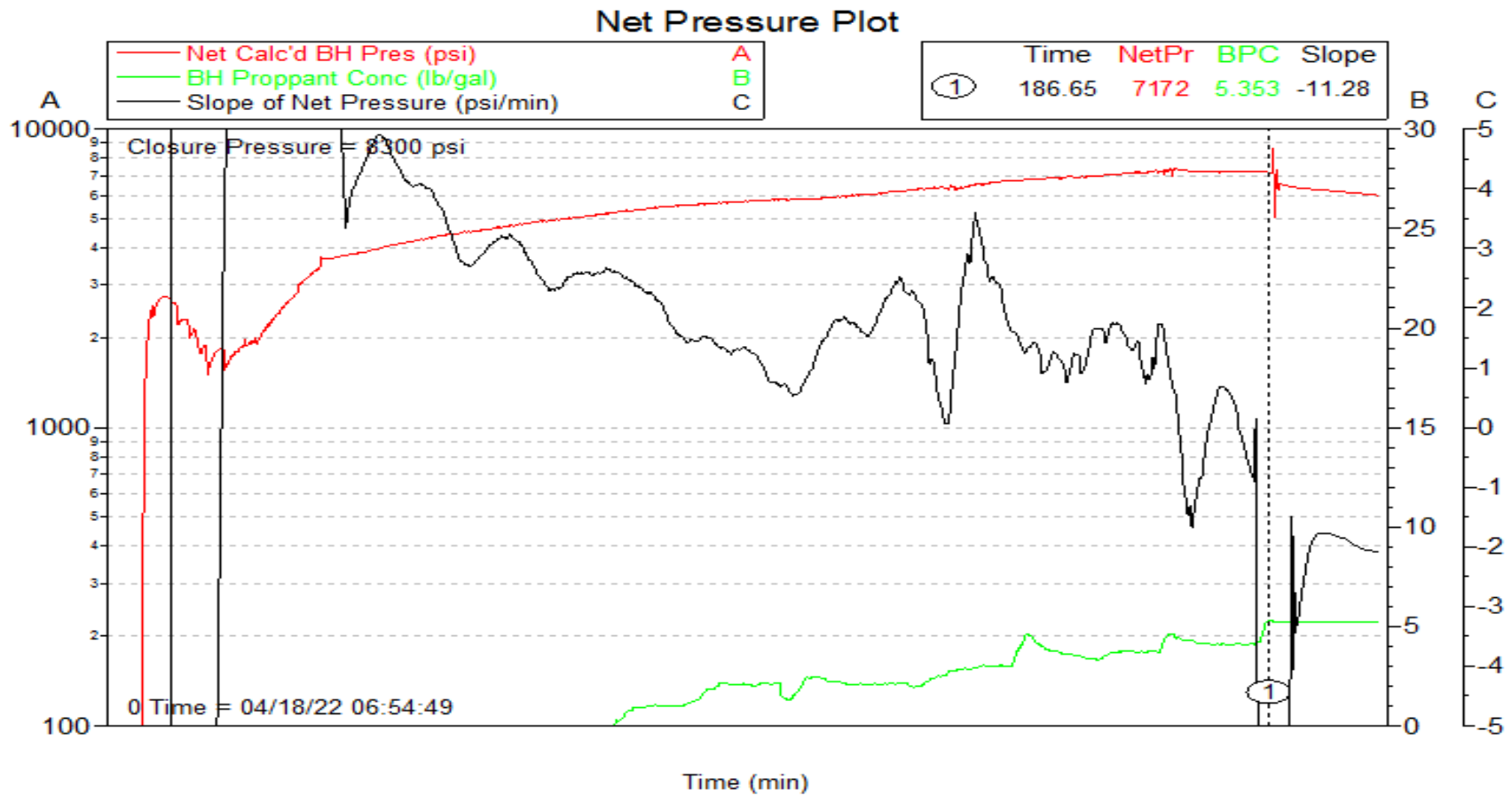
| Global Event Log                      |             |             |
|---------------------------------------|-------------|-------------|
| Intersection                          | Treating Pr | Backside Pr |
| 2 Global Kickout System Pressure Test | 122.3       | 32.42       |
| 3 Local Kickout System Pressure Test  | 1910        | 32.24       |
| 4 Treating Lines Pressure Test        | 12822       | 2336        |
| 5 Backside Line Pressure Test         | 95.59       | 3524        |
| 6 Treating Lines PRV                  | 11763       | 3520        |
| 7 Backside Line PRV                   | 214.4       | 2496        |

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6.9 Main Frac Chemical Additives Job Summary



## 6.10 Net Pressure Plot



## 6.11 Main Frac Job Summary

